LEGALGUARD AI

Project Files Directory & Architecture Guide (File Index & Roles)

Project Name:	LegalGuard AI / LegalFlow AI	**Total Documented Files:**	59 Core Files
Document Purpose:	Viva Defense & Teacher Code Review	**Architecture:**	FastAPI + React 18 + Expo Mobile
Core Risk Engine:	21 Legal Risk Categories & Heuristics	**Scoring Method:**	Deterministic Deduplication Formula
Offline Local AI:	Ollama (Llama-3 / Qwen / Mistral)	**Generated Date:**	September 28, 2026

Is Document Ka Upyog (How to use this guide):
Yeh official technical reference document aapke LegalGuard AI project ke har ek component, service, API endpoint, aur frontend UI page ki poori jankari pradan karta hai. Agar teacher ya examiner pooche ki *'Yeh file kya kaam karti hai?'*, to aap is index se seedha uska exact purpose, underlying technology aur system role explain kar sakte hain.

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1. Backend Core Services & AI Engines

Contracts ko parse karne, legal risks dhundhne, deterministic score nikalne aur auto-redline karne ke main engines.

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
01	**agent_engine.py** `backend/app/services/agent_engine.py`	Python / Engine	**Legal Risk Detection Engine:** Project ka sabse main analysis engine. Yeh 6-stage pipeline chalta hai: candidate extraction, missed-risk deep scan, safe-clause validator, aur semantic deduplication. 21 legal risk domains (jaise uncapped liability, non-compete, perpetual duration) ko detect karta hai aur screen par highlight karne ke liye PDF bounding-box coordinates attach karta hai.
02	**risk_engine.py** `backend/app/services/risk_engine.py`	Python / Engine	**Mathematical Scoring Engine:** Contract ka 0 se 100 ke beech deterministic safety score calculate karta hai. Har unique risk category par exact penalty deduct karta hai (Critical: -25, High: -15, Medium: -8, Low: -2). Ek hi category ke multiple occurrences hone par penalty sirf 1 baar deduct karta hai, aur contract ko GREEN, YELLOW, ORANGE, ya RED me classify karta hai.
03	**docx_redliner.py** `backend/app/services/docx_redliner.py`	Python / Docx	**Automated Contract Redlining Engine:** Khatarnak ya ek-tarfa clauses ko red strikethrough (strikethrough) karta hai aur unki jagah fair, balanced counter-clauses (green text) insert karke ready-to-negotiate Microsoft Word (.docx) file generate karta hai.
04	**comparison_service.py** `backend/app/services/comparison_service.py`	Python / Diff	**Clause-Aligned Contract Diff Engine:** Do contracts (Version A vs Version B) ko compare karta hai. Clauses ko legal subject (Fees, Liability, Termination, Security) ke hisaab se match karke MODIFIED, ADDED, REMOVED, aur UNCHANGED me classify karta hai aur percentage similarity nikalta hai.
05	**timeline_service.py** `backend/app/services/timeline_service.py`	Python / Service	**Contract Timeline & Milestones Extractor:** Document me se saare critical dates, deadlines, notice windows (jaise 30 days notice), aur payment schedules ko nikal kar chronological order me timeline banata hai.
06	**pdf_parser.py** `backend/app/services/pdf_parser.py`	Python / PyMuPDF	**Vector PDF Text & Coordinate Extractor:** PyMuPDF (fitz) library ka use karke uploaded PDF files me se text, clean paragraphs, page numbers aur exact word coordinates (x0, y0, x1, y1) nikalta hai taaki frontend par red highlight boxes sahi jagah draw ho sakein.
07	**ocr_service.py** `backend/app/services/ocr_service.py`	Python / Tesseract	**Optical Character Recognition (OCR) Engine:** Agar user koi scanned PDF ya photo/image upload kare jisme direct text na ho, to yeh Tesseract OCR engine chala kar image ko readable text me convert karta hai.
08	**ollama_service.py** `backend/app/services/ollama_service.py`	Python / LLM	**Local LLM Provider:** Locally running Ollama instance (Llama-3, Qwen, Mistral) se connect karta hai. Cloud par data bheje bina 100% offline, privacy-safe tareeqe se contract Q&A; aur summaries generate karta hai.
09	**template_service.py** `backend/app/services/template_service.py`	Python / Template	**Legal Template Generator:** Dynamic user inputs (Party names, dates, amounts) lekar standard legal agreements (NDAs, MSAs, Consulting Contracts) ki clean .docx file build karta hai.
10	**ai_provider.py** `backend/app/services/ai_provider.py`	Python / Base	**AI Provider Abstraction Interface:** Base class jo multiple AI models ya fallback providers ko ek unified format me connect karne ki suvidha deti hai.

2. Backend API Routes, Core Config & Security

FastAPI server ke endpoints, JWT authentication, CORS aur database setup.

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
11	**main.py** `backend/app/main.py`	FastAPI / App	**FastAPI Application Entry Point:** Server ka main file jo FastAPI app initialize karta hai, CORS permissions set karta hai, database tables create karta hai, aur REST, Chat, aur WebSocket routers ko mount karta hai.
12	**routes.py** `backend/app/api/routes.py`	FastAPI / REST	**Main REST API Endpoints:** Document upload (`/upload`), analysis trigger (`/analyze`), analysis report fetch (`/analysis/{id}`), contract comparison (`/compare`), aur template generation (`/templates/generate`) ke routes handle karta hai.
13	**chat.py** `backend/app/api/chat.py`	FastAPI / Chat	**Multilingual Legal Chatbot API:** User ke contract questions ka jawab deta hai. Local Ollama LLM use karta hai aur agar model offline ho, to zero-downtime deterministic fallback engine se exact page reference ke saath jawab deta hai (6 languages support).
14	**websocket.py** `backend/app/api/websocket.py`	WebSocket	**Real-Time Live Streaming:** Large document analysis ke time frontend par live progress bar aur stage-by-stage status events bhejne ke liye WebSocket connection manage karta hai.
15	**config.py** `backend/app/core/config.py`	Pydantic Settings	**Application Configuration:** Environment variables (.env) ko load karta hai: Database URL, JWT Secret Key, Ollama Host URL, Upload file size limits, aur allowed CORS origins.
16	**database.py** `backend/app/core/database.py`	SQLAlchemy Async	**Database Engine & Session:** SQLite database (`legalflow.db`) ke sath asynchronous connection banata hai aur session lifecycle manage karta hai.
17	**security.py** `backend/app/core/security.py`	Security / JWT	**Password Hashing & JWT Auth:** Passwords ko bcrypt se secure hash karta hai, JWT access tokens banata hai, aur protected routes par user authentication verify karta hai.

3. Backend Data Models & Database Schemas

Database tables aur request/response validation schemas.

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
18	**db_models.py** `backend/app/models/db_models.py`	SQLAlchemy ORM	**Relational Database Tables:** SQLite me tables define karta hai: `User`, `Document` (uploaded files), `DocumentPage` (page text & OCR data), `Clause` (extracted paragraphs), `RiskFinding` (detected legal risks), `AnalysisReport`, `ChatSession`, aur `ChatMessage`.
19	**schemas.py** `backend/app/models/schemas.py`	Pydantic Schemas	**Data Validation Schemas:** API request aur response ka structure validate karta hai: `UserCreate`, `TokenResponse`, `ClauseSchema`, `RiskFindingResponse`, `DocumentCompareResponse`, `BoundingBox`, aur `RiskLevelEnum`.

4. Backend Automated Test Suites

Deduplication accuracy, scoring logic, aur endpoints verify karne waale tests.

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
20	**test_deduplication_benchmarks.py** `backend/tests/test_deduplication_benchmarks.py`	Pytest / Async	**Deduplication Benchmark Tests:** Verify karta hai ki agar contract me multiple unlimited liability clauses hain (PDF 08), to score sirf ek baar deduct hoga, aur agar different legal effects hain (PDF 07 termination vs renewal), to dono alag-alag score honge.

21	**test_risk_engine.py** `backend/tests/test_risk_engine.py`	**Pytest / Unit**	**Risk Engine Scoring Tests:** Deterministic scoring formula (`Score = max(0, 100 - sum(weights))`) aur classifications (Green, Yellow, Orange, Red) ko mathematically test karta hai.
22	**test_chat_fallback.py** `backend/tests/test_chat_fallback.py`	**Pytest / Unit**	**Chatbot Offline Fallback Tests:** Check karta hai ki agar Ollama server band ho, tab bhi chatbot bina crash huye document risk report se accurate answer nikal kar deta hai ya nahi.
23	**test_auth.py** `backend/tests/test_auth.py`	**Pytest / API**	**Authentication Tests:** User registration, login token generation, guest session creation, aur duplicate email handling ko test karta hai.
24	**test_backend.py** `backend/tests/test_backend.py`	**Pytest / API**	**Server Health & Routes Tests:** FastAPI server health check endpoint (`/health`) aur core route responses ko verify karta hai.
25	**test_docx_redliner.py** `backend/tests/test_docx_redliner.py`	**Pytest / Docx**	**Word Document Redline Tests:** Generated redline .docx file ke andar paragraph styles, strikethrough runs aur replacement text colors ko verify karta hai.
26	**test_e2e_full.py** `backend/tests/test_e2e_full.py`	**Pytest / E2E**	**End-to-End Pipeline Test:** Document upload se lekar OCR, risk analysis, score calculation aur report generation ka poora flow ek saath test karta hai.

5. Frontend Web Application Pages

User interface ke main screens (React 18 + Tailwind CSS).

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
27	**Login.jsx** `frontend/src/pages/Login.jsx`	**React / Page**	**User Login Screen:** Email aur Password fields, Show/Hide password toggle button, client-side validation, aur error messaging ke sath clean login page jo authenticate karke Dashboard open karta hai.
28	**Register.jsx** `frontend/src/pages/Register.jsx`	**React / Page**	**Account Registration Screen:** Naye user ke liye Full Name, Email, Password fields ke saath registration form. Input validation aur successful registration ke baad direct Dashboard navigation handle karta hai.
29	**Dashboard.jsx** `frontend/src/pages/Dashboard.jsx`	**React / Page**	**Main Dashboard:** Upload kiye gaye saare contracts ki list, har contract ka safety score badge (0-100), classification color, total detected risks, aur quick action buttons (View, Delete, Re-analyze) dikhata hai.
30	**Upload.jsx** `frontend/src/pages/Upload.jsx`	**React / Page**	**Document Upload Screen:** Drag-and-drop file upload interface (PDF, DOCX, TXT support) jisme live upload progress aur processing animation chalti hai.
31	**DocumentDetail.jsx** `frontend/src/pages/DocumentDetail.jsx`	**React / Page**	**Split-Screen Contract Inspector:** Left side par real PDF viewer aur right side par Risk Findings Sidebar, Score Gauge meter, Redline counter-clauses aur Multilingual Chatbot drawer ko ek sath render karta hai.
32	**Compare.jsx** `frontend/src/pages/Compare.jsx`	**React / Page**	**Contract Version Comparison Screen:** Do contract versions ko side-by-side compare karta hai. Badle huye clauses ko Yellow (Modified), nayi clauses ko Green (Added), aur unchanged clauses ko Grey me highlight karta hai.
33	**Templates.jsx** `frontend/src/pages/Templates.jsx`	**React / Page**	**Draft Generator Screen:** Legal templates (NDA, MSA, Lease) select karke form fill karne aur 1-click me ready .docx draft download karne ki screen.
34	**Privacy.jsx** `frontend/src/pages/Privacy.jsx`	**React / Page**	**Privacy Center Screen:** Explains zero cloud telemetry, local on-premise AI processing, and institutional data privacy compliance.

6. Frontend Components, State & Services

Reusable UI components, state management aur API communication.

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
35	**Navbar.jsx** `frontend/src/components/Navbar.jsx`	React / Component	**Top Navigation Header:** LegalFlow AI branding, navigation links (Dashboard, Upload, Drafts, Compare, Privacy), logged-in user profile indicator, aur Logout button render karta hai.
36	**PDFViewer.jsx** `frontend/src/components/PDFViewer.jsx`	React / PDF.js	**Interactive PDF Renderer:** PDF.js ka use karke contract pages ko display karta hai, zoom in/out manage karta hai, aur risky clauses ke upar visual red highlight bounding boxes draw karta hai.
37	**RiskSidebar.jsx** `frontend/src/components/RiskSidebar.jsx`	React / Component	**Risk Findings Sidebar:** Detected legal risks ko severity ke hisaab se cards me dikhata hai. Risk ka legal effect, simple summary, aur suggested counter-amendment provide karta hai.
38	**ScoreGauge.jsx** `frontend/src/components/ScoreGauge.jsx`	React / SVG	**Visual Safety Score Gauge:** SVG-based semi-circular meter jo contract ka overall safety score (0 to 100) aur dynamic color classification (Green/Yellow/Orange/Red) visually animate karta hai.
39	**ChatDrawer.jsx** `frontend/src/components/ChatDrawer.jsx`	React / Component	**Multilingual AI Chatbot Drawer:** Slide-out chat widget jisme user contract ke baare me sawal pooch sakta hai. Language dropdown se Hindi, English, Marathi etc. select kar sakta hai.
40	**AuthContext.jsx** `frontend/src/context/AuthContext.jsx`	React Context	**Authentication State Management:** Global auth state manage karta hai: login, register, guest login, logout routines, aur localStorage me session persistence maintain karta hai.
41	**api.js** `frontend/src/services/api.js`	Axios Client	**Centralized API Client:** Axios instance jo automatically har request me JWT Bearer token attach karta hai aur backend ke saare REST endpoints ko frontend functions ke roop me export karta hai.
42	**App.jsx** `frontend/src/App.jsx`	React Router	**App Component & Routing:** Client-side routes define karta hai aur `ProtectedRoute` component ke zariye bina login kiye dashboard access hone se rokta hai.
43	**main.jsx** `frontend/src/main.jsx`	React Bootstrap	**Frontend Entry Point:** React 18 application ko DOM ke `#root` element me mount karta hai.
44	**vite.config.js** `frontend/vite.config.js`	Vite Config	**Vite Dev Server & Proxy:** Port 3000 par frontend host karta hai aur `/api` aur `/ws` requests ko backend port 8080 par proxy karta hai.

7. Mobile Application (React Native / Expo)

Mobile devices ke liye lightweight contract review interface.

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
45	**App.js** `mobile/App.js`	React Native	**Mobile App Entry Point:** Mobile screen layout, responsive document summary cards, aur mobile dashboard navigation render karta hai.
46	**app.json** `mobile/app.json`	Expo Config	**Expo Mobile Configuration:** App name, app icon, splash screen, package name aur Android orientation settings store karta hai.
47	**package.json** `mobile/package.json`	JSON Config	**Mobile Dependencies:** React Native, Expo SDK, vector icons aur mobile navigation packages define karta hai.

8. DevOps, Deployment & Startup Scripts

Project ko 1-click me chalane aur Docker containerization ke scripts.

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
48	**start.bat** `./start.bat`	Windows Batch	**Master 1-Click Launch Script:** Windows par double-click karne se ek sath do separate windows me Backend (Port 8080) aur Frontend (Port 3000) dono ko automatically start kar deta hai.
49	**start_backend.bat** `./start_backend.bat`	Windows Batch	**Backend Runner Shortcut:** Sirf FastAPI backend server (`uvicorn app.main:app`) ko port 8080 par launch karta hai.
50	**start_frontend.bat** `./start_frontend.bat`	Windows Batch	**Frontend Runner Shortcut:** Vite server ko Node ke zariye port 3000 par direct launch karta hai bina npm issues ke.
51	**start.ps1** `./start.ps1`	PowerShell Script	**PowerShell Environment Starter:** Environment check karke backend aur frontend processes ko PowerShell terminal me launch karta hai.
52	**docker-compose.yml** `./docker-compose.yml`	Docker Compose	**Multi-Container Orchestration:** Backend, Frontend aur Database ko single command (`docker-compose up`) se cloud ya Linux server par deploy karne ki config.
53	**backend/Dockerfile** `backend/Dockerfile`	Docker Blueprint	**Backend Docker Image:** Python 3.11 environment, Tesseract OCR system packages aur Python requirements install karke backend container banata hai.
54	**frontend/Dockerfile** `frontend/Dockerfile`	Docker Blueprint	**Frontend Docker Image:** Node.js environment me React build banata hai aur Nginx server ke zariye web UI serve karta hai.

9. Project Documentation & Academic Specifications

College report, viva examination aur system architecture files.

#	File Name & Location	Technology	Vo File Kya Kaam Kar Rahi Hai (Detailed Role)
55	**README.md** `./README.md`	Documentation	**Project Master Guide:** Poore system ka overview, architectural diagrams, quick start commands, API table, aur technology stack documentation.
56	**ARCHITECTURE.md** `docs/ARCHITECTURE.md`	Technical Spec	**System Architecture Document:** High-level layer diagrams: Presentation Layer (React), API Layer (FastAPI), Intelligence Layer (Risk & Ollama), Storage Layer (SQLite).
57	**SRS.md** `docs/SRS.md`	Software Spec	**Software Requirements Specification (SRS):** College project report ke liye functional requirements, non-functional performance benchmarks, aur hardware requirements.
58	**VIVA_QUESTIONS.md** `docs/VIVA_QUESTIONS.md`	Viva Prep	**Teacher Viva Defense Guide:** Examiners ke top expected questions, theoretical concepts (Deduplication, Risk Scoring, OCR vs Vector), aur model answers.
59	**LITERATURE_SURVEY.md** `./LITERATURE_SURVEY.md`	Academic Research	**Literature Survey & Comparison:** Existing legal tech tools (LegalBERT, Contract-Review) ka comparative analysis aur LegalGuard AI ke advantages.